ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
JAIPUR CENTER
Academic Session : 2021-22

SRGBT

for
JEE (MAIN+ADVANCED) 2022
ENTHUSIAST COURSE

Date : 26-09-2021 | Time : 3.00 Hr | CLASS-XII

Student's Form No. : ..... Batch : .....

Student's Name : .....

INSTRUCTIONS

1. **PART-1 : Physics** : **(Q. No. 1 to 10)** : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 11 to 20) : **Numerical Type Questions : Marking Scheme (+3, 0)**
2. **PART-2 : Chemistry** : **(Q. No. 21 to 30)** : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 31 to 40) : **Numerical Type Questions : Marking Scheme (+3, 0)**
3. **PART-3 : Mathematics** : **(Q. No. 41 to 50)** : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 51 to 60) : **Numerical Type Questions : Marking Scheme (+3, 0)**

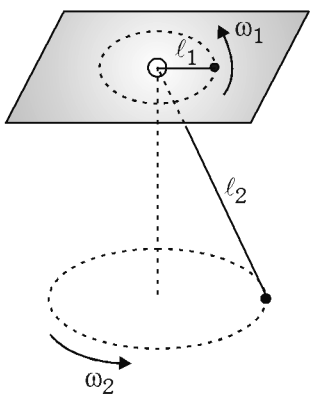
PART-1 : PHYSICS

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

1. To one end of a light inextensible string of length ℓ is attached a particle of mass m resting on a smooth horizontal a particle of mass m resting on a smooth horizontal table. The string passes through a smooth hole in the table and to its other end is attached a small particle of same mass. The system is set in rotation as shown in figure with constant angular speed. Choose the correct option(s):



(A) $\frac{\ell_1}{\ell_2} = \frac{\omega_2^2}{\omega_1^2}$

(B) $\frac{\ell_1}{\ell_2} = \frac{\omega_1^2}{\omega_2^2}$

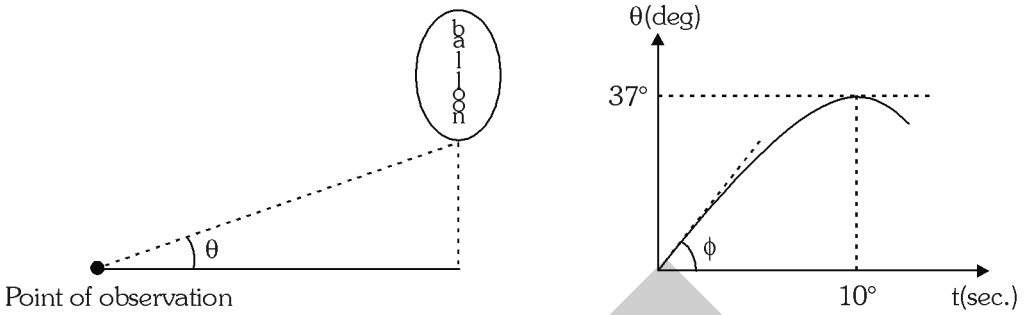
(C) The motion is possible if, $\frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} < \frac{\ell}{g}$

(D) The motion is possible if, $\frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} > \frac{\ell}{g}$

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For Q.no. (2 & 3)

Wind velocities at various height can be investigated using balloons with constant ascending velocity. A graph between angle of elevation (θ) against time is shown in the diagram by observing such a balloon. The balloon was released at distance 900 m from the point of observation from the same level and it seemed to be rising directly upwards. Wind velocity near the ground was zero. Wind velocity near the ground was zero. Wind velocity is horizontal at each level.



2. Choose the correct statement(s) :

(A) During the journey of the balloon its distance from point of observation always dec reases.

(B) During the journey of the balloon its distance from point of observation increases.

(C) Angular velocity of balloon with respect to point of observation is 3.7 deg/sec. at t = 10 sec.

(D) Angular velocity of balloon with respect to point of observation is zero at t = 10 sec.

3. If slope of tangent of graph at t = 0, i.e., $\tan\phi = \frac{6}{\pi}\text{deg/sec.}$, then :

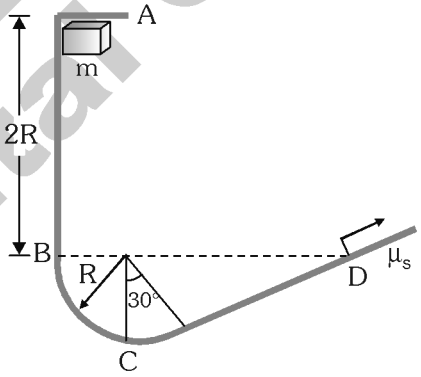
(A) Angular velocity of balloon about point of observation is zero at t = 0.

(B) Velocity of ascending of balloon is 30 m/s

(C) Velocity of wind at height 300 m is 40 m/s

(D) Height of balloon at t = 10 sec is 40 m.

4. A block of mass m is released on a track shown in figure. The track is smooth from A to D and rough from point D onwards. Based on above information, answer the following questions.



(A) The normal force N_B exerted by the track just after crossing point B is 4 mg

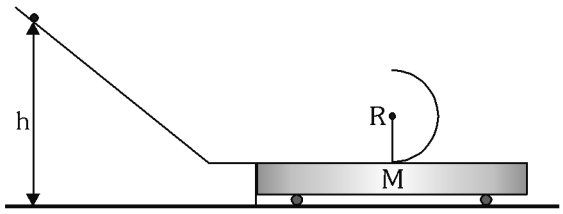
(B) The normal force N_C at C is 7 mg

(C) The normal force N_B exerted by the track just after crossing point B is 7 mg

(D) The distance travelled along the incline beyond point D where the block stops is $\frac{4R}{\sqrt{3}\mu_k + 1}$

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5. A small object of mass m = 1 kg is released from rest at the top of an inclined plane that connects to a horizontal plane without an edge. It slides onto a cart of mass M that has a semi-cylindrical surface of radius R = 0.36 m fixed to the middle of it, as shown in the figure.



The small object reaches the topmost point of the semi-cylinder and stops there. In continues to move vertically with free fall and hits the cart exactly at the edge. All friction can be neglected.
Choose correct option(s) :
(A) the minimum possible length of the cart is 1.44 m
(B) the mass of the cart is 2.56 kg
(C) height h was the small object released is 1.81 m
(D) the mass of the cart is 5 kg

6. A small object of mass m is suspended by a thread of length ℓ and released when thread is horizontal.

(A) Velocity of the object when vertical component of its velocity is maximum $\sqrt{\frac{2}{\sqrt{3}}g\ell}$.

(B) Tension in the thread when vertical component of velocity is maximum $\sqrt{3}\,mg$.

(C) Ratio of maximum vertical velocity to maximum horizontal velocity is $\sqrt{\frac{2}{3\sqrt{3}}}$.

(D) Ratio of maximum vertical velocity to maximum horizontal velocity is $\sqrt{\frac{2}{\sqrt{3}}}$.

7. A particle is projected in air from origin with speed u and at an angle θ with positive x-axis. At its maximum height its trajectory has radius of curvature R and centre of curvature has co-ordinate $\left(20m, \frac{70}{3}m\right)$. Choose the correct option(s). $(\vec{g} = -10 m/s^2\,\hat{j})$

(A) Equation of trajectory of the projectile motion is $y = 3x - \frac{3x^2}{40}$.

(B) The maximum height of the projectile is 30 m.

(C) The value of u is $\sqrt{\frac{2000}{3}}\ m/s$.

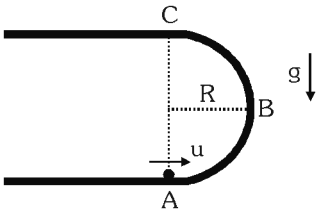
(D) The value of θ is $\tan^{-1}\left(\frac{3}{\sqrt{10}}\right)$.

8. A particle moves along a circular path of radius 3m about a stationary axis according to the law $\theta = (t^3 - 3t^2)\pi$ radian. Then which of the following is/are correct.

(A) The magnitude of mean value of the angular velocity over the time interval between t = 0 and the time when the particle comes to rest, is 2π rad/sec.
(B) The magnitude of average speed over the time interval between t = 0 to t = 3 sec is 8π m/sec.
(C) The distance travelled by the particle between the time interval t = 1 to t = 4 sec, is 66π meter.
(D) The distance travelled by the particle between the time interval t = 1 to t = 4 sec, is 60π meter.

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9. Figure shows a smooth fixed track with two straight parts connected by a semicircular arc ABC. The shown track is placed in a vertical plane. A small ball is given an initial speed 'u' along the track as shown. The ball loses contact with the track somewhere in the part BC and collides with the track at A during subsequent motion. Now answer the following questions.



(A) the value of 'u' is $\sqrt{\frac{7}{2}Rg}$

(B) time after which it collides at 'A' after losing the contact with the shown track, is $\sqrt{\frac{6R}{g}}$

(C) the value of 'u' is $\sqrt{\frac{9}{2}Rg}$

(D) The time after which it collides at 'A' after losing the contact with the shown track, is $\sqrt{\frac{3R}{g}}$

10. The potential energy function of a particle is given by $U(r) = \frac{A}{2r^2} - \frac{B}{3r}$, where A and B are constants and r is the radial distance from the centre of the force. Choose the correct option(s) :

(A) The equilibrium distance will be $r_0 = \frac{2A}{B}$

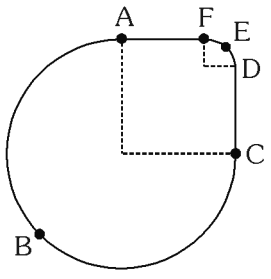
(B) The equilibrium distance will be $r_0 = \frac{3A}{B}$

(C) If the total energy of the particle is $\frac{B^2}{6A}$, then its radial velocity will vanish at $\frac{r_0}{3}$, where r_0 is the equilibrium distance.

(D) If the total energy of the particle is $\frac{B^2}{6A}$, then its radial velocity will vanish at $\frac{r_0}{2}$, where r_0 is the equilibrium distance.

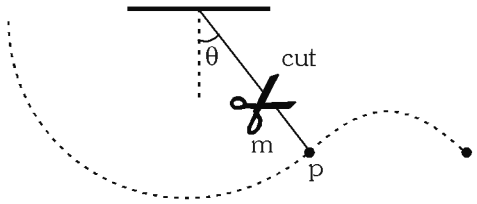
NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

11. A racing track ABCDEFA is such that part ABC is circular of radius 400 m, part CD and FA are straight of length 300 m and part EF is circular of radius 100 m. The coefficient of friction between track and car tyres is 0.1. A racing car always moves on this track with uniform speed. The minimum time (in sec.) that a racing car can take to complete one complete lap is _____ .

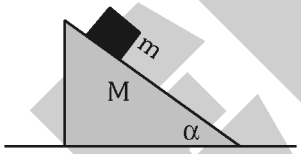


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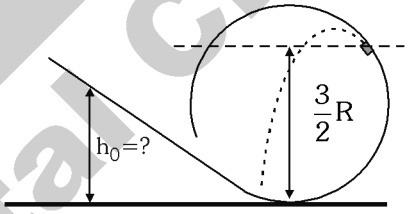
12. Consider a pendulum, consisting of a massless string with a mass on the end. The mass is held with the string horizontal, and then released. The mass swings down, and then on its way back up, the string is cut at point P when it makes an angle of θ with the vertical. What should be the value of $111 \times \cot^2 \theta$, so that the mass travels largest horizontal distance from P by the time it return to the height it had when the string was cut ?
($g = 10\ m/s^2$)



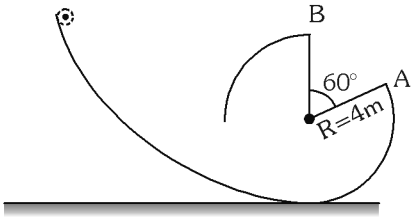
13. A wedge of mass M is kept at rest on an horizontal surface, and a block of mass m is kept on the wedge at the height h from the surface. the angle of the wedgeis α, see Fig. There is no friction neither between the block and wedge nor between the surface and the wedge. The system is released into a free motion. Find the time (in sec.) needed for the block to reach the surface. (Given M = 9m, h = 400 m, $\tan \alpha = \frac{1}{\sqrt{2}}$)



14. A tangentially attached slope leads to a circular match-box track with radius R = 32 cm set in a vertical plane. The toy car starts from rest at the top of the slope, runs down the slope and detaches from the track at height h = 3/2 R measured from the bottom. Find the $\left(\frac{\text{maximum height}}{4}\right)$ (in cm) reached by it after it reaches the bottom of the track. (Assume that the toy car is point-like, neglect drag and friction).



15. As shown in the figure, a ball is released from a certain height which has to perform circular motion on the vertical track of radius 4 m. The track is absent between points A and B. Calculate the height (in cm) from where the ball has to be released so that it will reach at highest point B of the circular track.



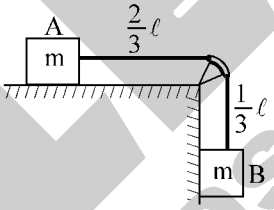
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16. In a park there are three concentric circular running tracks. Radius of 2nd track is double of first and of 3rd track is triple of first. Three runners are running on these tracks with constant speed. When the runner in the first track completes one round, the runner in 2nd has completed half round and the runner in third track has completed quarter round. If the accelerations of the runners are in ratio $\alpha : \beta : \gamma$, where α, β, and γ are least integers, then find the value of $4(\alpha + \beta + \gamma)$.

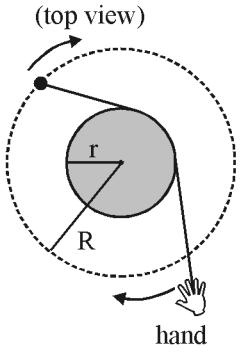
17. Two identical cubes of mass 1.5M each are kept almost touching the faces on a smooth horizontal surface. On top of them we neatly placed a smooth ball of mass M, which begins to move vertically downwards, pushing the cubes sideways. The initial velocity of the ball is negligible. Radius of the ball is 5m, side of cube 2.5m. If velocity of ball just before impact with ground is V. Find $\frac{3V^2}{4}$.

18. A rod is rotating with an angular velocity of $3\pi t^3$ rad/s. The angle rotated by the rod from t = 0 to the time when it has an angular velocity of 24π rad/s is given by $\alpha\pi$ radian. Fill $\frac{\alpha}{2}$ in OMR sheet.

19. Two objects A and B of mass m are connected with a uniform rope of length ℓ = 0.3 m and mass m as shown in the figure. The rope can slide in a tube which forms a quarter of a circle, and which is fixed to the edge of the table. Initially 1/3 of the rope hangs vertically. Friction is negligible. If the system is released from rest, and B descends a distance of $\ell/3$, what is the speed (in m/s) of B?



20. A mass m = 1kg, which is free to move on a horizontal frictionless surface, is attached to one end of a massless string that wraps partially around a frictionless vertical pole of radius r (see the top view in figure). You hold on to the other end of the string. At t = 0, the mass has speed v_0 in the tangential direction along the dotted circle of radius R shown. Your task is to pull on the string so that the mass keeps moving along the circular path of radius R. If at any instant 't' force applied by hand on string is f_0 = 2N then find out centripetal acceleration (in m/s^2) of mass. (Given : $R = \frac{2r}{\sqrt{3}}$)



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PART-2 : CHEMISTRY

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

21. Select **CORRECT** statement(s).

(A) O_3 and O^{2-} both are diamagnetic

(B) Out of O_2, O_2^+ and O_2^-, O_2 have maximum magnetic moment (spin only)

(C) Out of O_2, O_2^+ and O_3, O_2^+ have least O–O bond length

(D) Out of O_2, O_2^{2+} and O_2^{2-}, only O_2 is paramagnetic.

22. Select the option(s) having correct order of the properties involved.

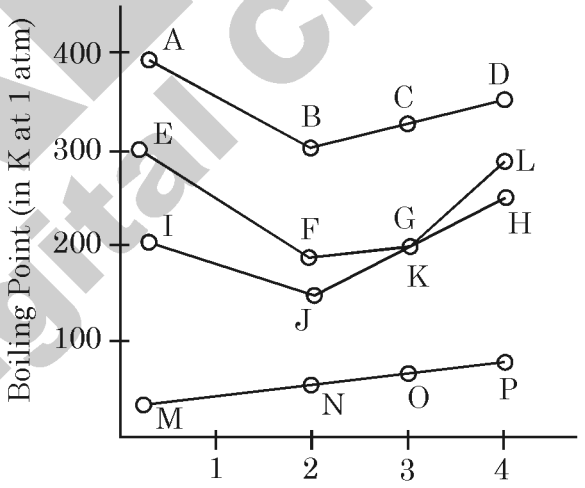
(A) Size in gas phase : $Al^{3+} < Ga^{3+} < H^-$

(B) IE_1 values : Sb < Te

(C) At. size : Ag > Hg

(D) IE_1 values : Bi > Sb

23. Consider the boiling point of some molecular hydrides and monoatomic gases



Molecular Hydrides/Monoatomic gas

According to the given information the correct match are :

(A) A to D compounds are H_2O, H_2S, H_2Se, H_2Te respectively

(B) E to H compounds are HF, HCl, HBr, HI respectively

(C) I to L compounds are CH_4, SiH_4, GeH_4, SnH_4 respectively

(D) M to P are Ne, Ar, Kr, Xe respectively

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24. One mole of 'F' atoms are ionised to F^- the energy released is 'X' Joules. Then the incorrect statements:
(A) X Joules is sufficient to ionise 1 mole of gaseous Cl into Cl^+.
(B) X Joules is sufficient to ionise 1 mole of gaseous F into F^+.
(C) X Joules is sufficient to ionise 1 mole of gaseous F into F^+ as well as 1 mole of gaseous Cl into Cl^+.
(D) Less than 1 mole of gaseous F or Cl will be ionised to F^+ or Cl^+.

25. Find correct match.

	Molecule	**Nature of bridge bond**	**Planar/Non–planar**
(A)	Al_2Cl_6	$3C–4e^-$	NP
(B)	I_2Cl_6	$3C–4e^-$	P
(C)	$Al_2(CH_3)_6$	$3C–2e^-$	NP
(D)	$(BeCl_2)_n$	$3C–4e^-$	NP

26. X, Y and Z are three consecutive elements of second period whose I.P. data (in eV) are available. By using these data specify incorrect statements regarding X, Y and Z.

Element	IP_1	IP_2	IP_3	IP_4
X	5.39	75.6	122.4	Not possible
Y	8.2	25.1	37.9	259.2
Z	9.3	18.2	217.6	363.7

(A) $X_{(g)} + e^- \longrightarrow A^-_{(g)}$ ΔH = positive
(B) $Z_Y > Z_Z$ (Z is atomic number)
(C) Oxide of 'Y' is amphoteric
(D) X^{+1} is stable than X^{2+}

27. Which of the following statements are correct ?
(A) Vapour pressure of H_2O is more than H_2Te at given temperature.
(B) In both N_2O_5 and N_2O_4 all N–O bond lengths are equivalent.
(C) In both crystalline $NaHCO_3$ and $KHCO_3$, HCO_3^- forms different type of association of anion through hydrogen bond.
(D) Amongst B_2, C_2, N_2^- and O_2, N_2^- and O_2 on further ionization (losing single electron) forms thermodynamically more stable species.

28. Select the option containing correct order of the parameters given
(A) Increasing order of acidic strength : Na_2O, MgO, CO_2, SO_3
(B) Decreasing order of ionic radius : K^+, Na^+, Li^+, Mg^{2+}
(C) Decreasing order of boiling points : H_2O, HF, NH_3
(D) Decreasing order of boiling points : He, Ne, Ar, Xe

29. X and Y are second period elements. They form two compounds XF_3 and YH_3 respectively. XF_3 is resonance stabilized while YH_3 is not. Select the incorrect statement.
(A) X–F bond distance in XF_3 is smaller than that in XF_4^-.
(B) X–F bond in XF_3 is larger than that in $XF_3^- – YH_3^+$.
(C) XF_3 does not have back bonding.
(D) YH_3 is a Lewis acid.

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30. Which of the following statements are **correct** :

(A) IE_2 of Cu > IE_2 of Zn where as IE_1 of Cu < IE_1 of Zn.

(B) $|\Delta H_{EG}|$ of S > $|\Delta H_{EG}|$ of O.

(C) Reducing nature of Li in aqueous solution is more than that of Na.

(D) There is a substantial contraction in size observed on moving from 3rd group to 4th group in 6th period of periodic table.

NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

31. Consider following molecular species :

XeF_4, ClO_4^-, XeO_3, IF_4^+, ClF_3, SF_6, SF_4, PCl_6^-, XeO_6^{4-}, $S_3O_6^{2-}$

and calculate $\frac{(x+y)}{z}$.

x = Total number of molecular species in which all bond lengths are not equal
y = Total number of species in which 'd' orbital which having zero nodal plane participate in hybridisation
z = total number of planar species

32. How many comparison(s) are **INCORRECT** among given ?

(i) H > Li (ionisation energy)
(ii) Li > Be (size)
(iii) Na > Rb (size)
(iv) O > N (size)
(v) S > P (ionisation energy)
(vi) C > O (size)
(vii) B > Al (size)
(viii) F > Cl (ionisation energy)
(ix) N > O (ionisation energy)

33. How many of the following has underlined atom in sp^2 hybridisation ?
$\underline{Si}O_2$, $\underline{N}(SiH_3)_3$, $\underline{B}F_3$, $\underline{Al}Cl_3.6H_2O$, $H_3\underline{O}^{\oplus}$, $B_3\underline{N}_3H_6$, $\underline{S}O_2$, $\underline{N}O_2$

34. Assuming screening by K shell electrons to be same in Li & Be. Calculate $I.E_1$ (in eV) of Li if IE_2 of Be is 13.6×1.44 eV/atom.

35. Calculate the bond length of C–X bond, if C–C bond length is 1.54 Å, X–X bond length is 1.00 Å and electronegativity values of C and X are 2.0 and 3.0 respectively

36. From the following data calculate % ionic character in the bond A-B.
[1 eV/atom = 96 KJ/mole]

ΔH_{EG_1} of A^{+1} = –960 KJ/mole
IE_1 of A^- = –1.6 eV/atom

ΔH_{EG_1} of B = –192 KJ/mole
IE of B = 3.6 eV/atom

37. (i) Number of π–bonds in C_2 molecule (according to MOT) = x.
(ii) Number of ABMO electrons in O_2^- ion = y
(iii) Number of ABMO having energy less than $\sigma 2p_z$ in F_2 molecule = z.

Specify : $\frac{2x+4y-z}{600} = ?$

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38. The number of correct trends is/are :

(i) AgF > AgCl > AgBr > AgI (Solubility in water)

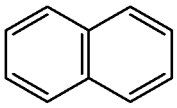
(ii) $TiO_2 < SiO_2$ (Polarising power)

(iii) $O_2^- > O_2^{2-}$ (Ionic radii)

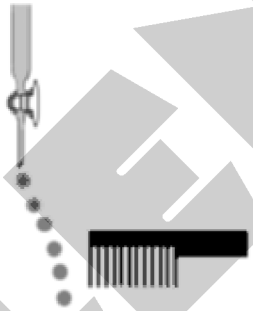
(iv) Cu > Ag < Au (Ionisation energy)

(v) N < O < F < Ne (Z_{eff})

(vi) $La^{3+} > Nd^{3+} > Gd^{3+} > Ho^{3+} > Lu^{3+}$ (Ionic radii)

39. Consider the following compounds in liquid form  , SO_2, $SiCl_4$, S_2, PCl_2F_3, O_3, CrO_5, $C_{12}O_9$, Cl_2.

When a charged comb is brought near their following steam how many of them deflection as per the following figure ?



40. Lattice energy of how many ionic compounds are greater than NaCl.
NaF, KCl, MgO, MgF_2, $A\ell N$, CsBr, NaI, Na_2O

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PART-3 : MATHEMATICS

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

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Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

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41. Let a, b, c be the sides of a triangle such that $\frac{bc}{b+c}+\frac{ca}{c+a}+\frac{ab}{a+b}=s$

where s is semi – perimeter, then the triangle is

(A) equilateral (B) isosceles but not right angled

(C) right angled but not isosceles (D) obtuse angled

42. If $3\sin\alpha\sin\beta+4\cos\beta+5\sin\beta\cos\alpha=5\sqrt{2}$, where $\alpha,\beta\in R$, then

(A) $\tan\alpha=\frac{3}{5}$ (B) $\tan\alpha=\frac{5}{3}$

(C) $\cos\alpha=\frac{3}{\sqrt{34}}$ (D) $\tan\beta=\frac{\sqrt{34}}{4}$

43. For a ΔABC with integer side lengths and $\cos A=\frac{11}{16}, \cos B=\frac{7}{8}, \cos C=-\frac{1}{4}$, which of the following statement(s) is/are true?

(A) least possible perimeter of triangle ABC is 27 (B) least possible perimeter of triangle ABC is 9

(C) least area of the triangle ABC is $\frac{3\sqrt{15}}{4}$ (D) None of these

44. If $\tan^2 x+\tan^2 y+\cot^2(x+y)=1, x, y\in\left(0,\frac{\pi}{4}\right)$, then :

(A) x can only be $\frac{\pi}{6}$ (B) y can only be $\frac{\pi}{6}$

(C) x can take more than 1 value (D) none of these

45. In ΔABC, P is any point inside a triangle such that area of ΔBPC, ΔAPC, ΔAPB are equal. Line AP cut BC at M, area of ΔPMC is 5 sq. unit then area of ΔABC can be :

(A) 20 sq. units (B) 25 sq. units (C) 30 sq. units (D) 10 sq. units

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46. In a ΔABC $\sin(A-B)=\frac{3}{5}$, $\sin A = 2\sin B$ then $\frac{a+b}{c}=$

(A) $\frac{4}{\sqrt{5}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\frac{3}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{5}}$

47. In a ΔABC, if $a = 2$, $B = 60°$, $C = 75°$, then

(A) $b = \sqrt{6}$ units (B) area of $\Delta ABC = \left(\frac{3+\sqrt{3}}{2}\right)$ sq. units.

(C) Length of angle bisector of $\angle B$ is 2 units (D) Circumradius of the triangle ABC is $\sqrt{2}$ units

48. $\sqrt{\cos 2x}+\sqrt{1+\sin 2x}=2\sqrt{\sin x+\cos x}$ if

(A) $\sin x+\cos x=0$ (B) $x = 2n\pi$

(C) $x = n\pi-\frac{\pi}{4}$ (D) $x = 2n\pi \pm \cos^{-1}\left(\frac{-1}{5}\right)$

49. If in a ΔABC, a, b, c are in A.P. then it is necessary that :

(A) $\frac{2}{3}<\frac{b}{c}<2$ (B) $\frac{1}{3}<\frac{b}{c}<\frac{2}{3}$ (C) $\frac{2}{3}<\frac{b}{a}<2$ (D) $\frac{1}{3}<\frac{b}{a}<\frac{2}{3}$

50. The solution set of $|\sin x| \le |\cos 2x|$ contains

(A) $\bigcup_{n\in I}\left\{\left[n\pi-\frac{\pi}{6}, n\pi+\frac{\pi}{6}\right]\right\}$ (B) $\bigcup_{n\in I}\left\{n\pi+\frac{\pi}{2}\right\}$

(C) $\bigcup_{n\in I}\left\{\left[n\pi-\frac{\pi}{8}, n\pi+\frac{\pi}{8}\right]\right\}$ (D) $\bigcup_{n\in I}\left\{\left[n\pi-\frac{\pi}{4}, n\pi+\frac{\pi}{4}\right]\right\}$

NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+3 , 0)

51. A triangle ABC has A – B = 120° and the circumradius is 8 times the in–radius. If cos C = a/8, then a =

52. The number of ordered pair (x,y) satisfying equation sinxcosy =1 and $x^2+y^2 \le 9\pi^2$ is m the find $\frac{78}{2m}$

53. Total number of ordered pairs (x, y) satisfying $x\left(\sin^2 x+\frac{1}{x^2}\right)=2\sin x.\sin^2 y$, where $x \in (-\pi,0)\cup(0,\pi)$ and $y \in [0,2\pi]$ is / are

54. The least degree of a polynomial with integer coefficient whose one of roots may be cos 12° is :

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55. If $x = \sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}$ and $y = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}$, then $\frac{5(x^2+y^2)}{2}$ is

56. ABCD is a trapezium. AB and CD are parallel and AD is perpendicular to them. Let AB = 4, CD = 3 and r be the radius of the circle which touches all the sides of trapezium them the value *of* [r] is (where [.] denotes greatest integer function)

57. Number of roots of the equation $|\sin\theta| + |\cos\theta| = \tan^2\theta + \cot^2\theta$

58. The sides a, b, c of ΔABC satisfy the relations $a + b - c = 2$ and $2ab - c^2 = 4$, if $x \in [0,1]$ then max value of $ax^2 + bx + c$ is :

59. In parallelogram ABCD, P is any point inside parallelogram such that sum of area of ΔAPD and ΔBPC is 3 sq. unit, then area of parallelogram ABCD is

60. The number of solutions of the equation $\tan x + \sec x = 2\cos x$ lying in the interval $[0, 2\pi]$ is :

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SRGBT-10

ENTHUSIAST COURSE

DATE : 26-09-2021

ANSWER KEY

PART-1 : PHYSICS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	BD	BC	ABD	ABC	ABC	ABC	AB	AB	BC
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	264	222	28.3	13.5	900	108	18.75	6	1	1

PART-2 : CHEMISTRY

Que.	21	22	23	24	25	26	27	28	29	30
Ans.	ABCD	AB	ABD	ABC	ABCD	AC	CD	ABC	BCD	ABCD
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	5.50	4	5	6.7	1.18	16.9	0.05	5	4	5

PART-3 : MATHEMATICS

Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	AD	BC	AB	C	C	ABCD	ABC	AC	AB
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	7	13	8	4	5	1	0	6	6	2